

Fall 2025, 4100 Problem Set 2, Hayden Fuller

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Given $A = \begin{bmatrix} 3 & -1 \\ 0 & 2 \end{bmatrix}$ show that the set $C = \{B \in \mathbb{R}^{2 \times 2} : AB = BA\}$ is a subspaces of $\mathbb{R}^{2 \times 2}$ and find a basis.
2. Let V be a vector space such that $\dim V = n$. Show that any set of n (non-zero) vectors, $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, satisfying $\mathbf{v}_i^T \mathbf{v}_j = 0$ for $i \neq j$ is a basis for V .
3. Let $\mathbb{P}_n$ be the space of all polynomials over $\mathbb{R}$ of degree at most n . Use the Rank-Nullity theorem to show that the subset $S = \{g \in \mathbb{P}_n : f' + 4f = g \text{ for some } f \in \mathbb{P}_n\}$ is a subspace of $\mathbb{P}_n$ of dimension $n + 1$.
4. Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times l}$ such that $AB\mathbf{x} = \mathbf{0}$ for all $\mathbf{x} \in \mathbb{R}^l$. Prove that $\text{rank}(A) + \text{rank}(B) \leq n$ using the Rank-Nullity theorem.
5. Let V be a vector space with basis $\{\mathbf{v}_1, \dots, \mathbf{v}_{n-1}\}$ and let $T \in \mathcal{L}(V, V)$ be defined by $T\mathbf{v}_i = \mathbf{v}_{n-i}$. Show that $T^{-1} = T$ and that $T - I$ is not invertible. (Here $(T - I)\mathbf{x} = T\mathbf{x} - \mathbf{x}$)
6. Consider the block-diagonal matrix $A = \begin{bmatrix} B & 0 \\ 0 & C \end{bmatrix}$. Show that $\text{nullity}(A) = \text{nullity}(B) + \text{nullity}(C)$.